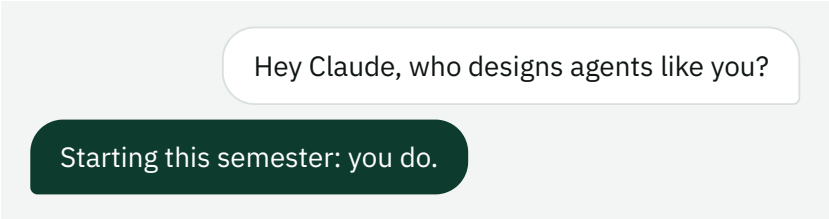
TS



STUDENT WORKBOOK · WINTER TERM 2026/27

Conversational Agent Design Lab

Department of Information Systems · University of Münster

VERSION 7 · 20.10.2026

TEAM _____

NAME _____

TOPIC _____

HOW TO USE THIS WORKBOOK

Seven steps, nine sessions, one chain of design knowledge

This workbook follows you through the semester. Each of the seven steps (E1 to E7) begins with an overview: what happens, what you should be able to do, the key concepts, your tasks, the deliverable, the exit check and what to read. Worksheets follow each step. Fill them in as a team; they are the raw material for your seminar paper.

The companion site holds the same content plus the Theory Lab: small simulated agents that let you try out the theories behind your design decisions, the case marketplace and the library.

	STEP	WEEKS	SESSIONS	RESULT
E1	Problem	KW 43–44	Kickoff	D1 Problem brief
E2	Requirements	KW 45–46	Character and problem space	D2 Character, objectives and requirements
E3	Design and build 1	KW 47–48	Vibe Coding 1	D3 Prototype v1 and feature trace
E4	Demonstrate	KW 49–50	User testing, Interim presentations	D4 Proof of concept (test round 1)
E5	Design and build 2	KW 51–01	Vibe Coding Advanced	Prototype v2, deployed
E6	Evaluate	KW 02	Team work with users	D5 Proof of value (test round 2)
E7	Abstract	KW 03–05	Design Knowledge Creation Video, Industry input: agents in practice, Final presentations	D6 Design principles and final submission

What you submit

Seminar paper. WI template, about 12 pages; design principles in the anatomy format

Artifact. Working, deployed agent; every feature traced to a requirement

Workbook. Worksheets and design journal with AI transparency statement; the more individual contribution

Presentations. Interim and final presentation

The weighting is not fixed yet. We publish it here and on LearnWeb as soon as it is.

Team



Dr. Timo Strohmann, Theory, design science, kickoff; Companion case



Amilia Fröhlich, Character workshop and user testing; FinBot and VIVA cases



Hans-Henning Näscher, Vibe coding and agent development; personal finance and Second Thought cases



Dr. Rangina Ahmad, Lead AI Agent Architect, Parloa; industry input in KW 03 or 04

Ground rules

- Supervision: your case supervisor is your first contact, after each session and whenever you need it.
- Feedback: we always welcome feedback, on the sessions, the material and the site. Tell us after a session, or write to your supervisor or to Timo.
- AI tools are welcome, including for code and writing. Log what you used in your design journal and submit an AI transparency statement (W10).
- Test users are people: informed consent, and no personal data in prompts or logs you cannot protect.

SEMESTER PLAN

Sessions and deadlines

	SESSION	WHEN	OWNER	BRING
1	Kickoff	Tue 20.10.2026, 13:00 to 16:00	Timo Strohmann	Curiosity; A laptop; One agent you use or used, and one thing that annoys you about it
2	Character and problem space	KW 45 (02.11 to 06.11), day open	Amilia Fröhlich	D1 Problem brief; Notes from at least three user conversations; One quote that surprised you
3	Vibe Coding 1	KW 47 (16.11 to 20.11), day open	Hans-Henning Näscher	D2 Character, objectives and requirements; A laptop with the tools Hans announces
4	User testing	KW 49 (30.11 to 04.12), day open	Amilia Fröhlich	D3 Prototype v1, running, and your feature trace (W5); Names of possible test users
5	Interim presentations	KW 50 (07.12 to 11.12), day open	All	Your talk; A running prototype; One requirement you now doubt
6	Vibe Coding Advanced	KW 51 (14.12 to 18.12) or KW 02 (11.01 to 15.01) (week open)	Hans-Henning Näscher	Your prototype and your open technical problems
7	Design Knowledge Creation Video	KW 02 (from 11.01.2027), self-paced	Timo Strohmann	Your feature trace (W5); Your round 2 results with the reasons why (W8)
8	Industry input: agents in practice	KW 03 or KW 04 (18.01 to 26.01), day open	Rangina Ahmad (Lead AI Agent Architect, Parloa)	Your draft design principles (W9); One question about building agents in practice
9	Final presentations	Wed 27.01.2027	All	Prototype v2; Your design principles

Deadlines

End of KW 44 (Sun 01.11)	D1 Problem brief
End of KW 46 (Sun 15.11)	D2 Character, objectives and requirements
End of KW 48 (Sun 29.11)	D3 Prototype v1 and feature trace
KW 50, with the interim presentation	D4 Proof of concept (test round 1)
End of KW 01 (Sun 10.01)	Prototype v2 deployed
End of KW 02 (Sun 17.01)	D5 Proof of value (test round 2)
Fri 05.02.2027	D6 Design principles and final submission
Fri 09.04.2027	WI 2027 Student Track (optional)

For now the plan shows calendar weeks (KW); the exact days follow after the kickoff. The current plan is always on the companion site.

CASES

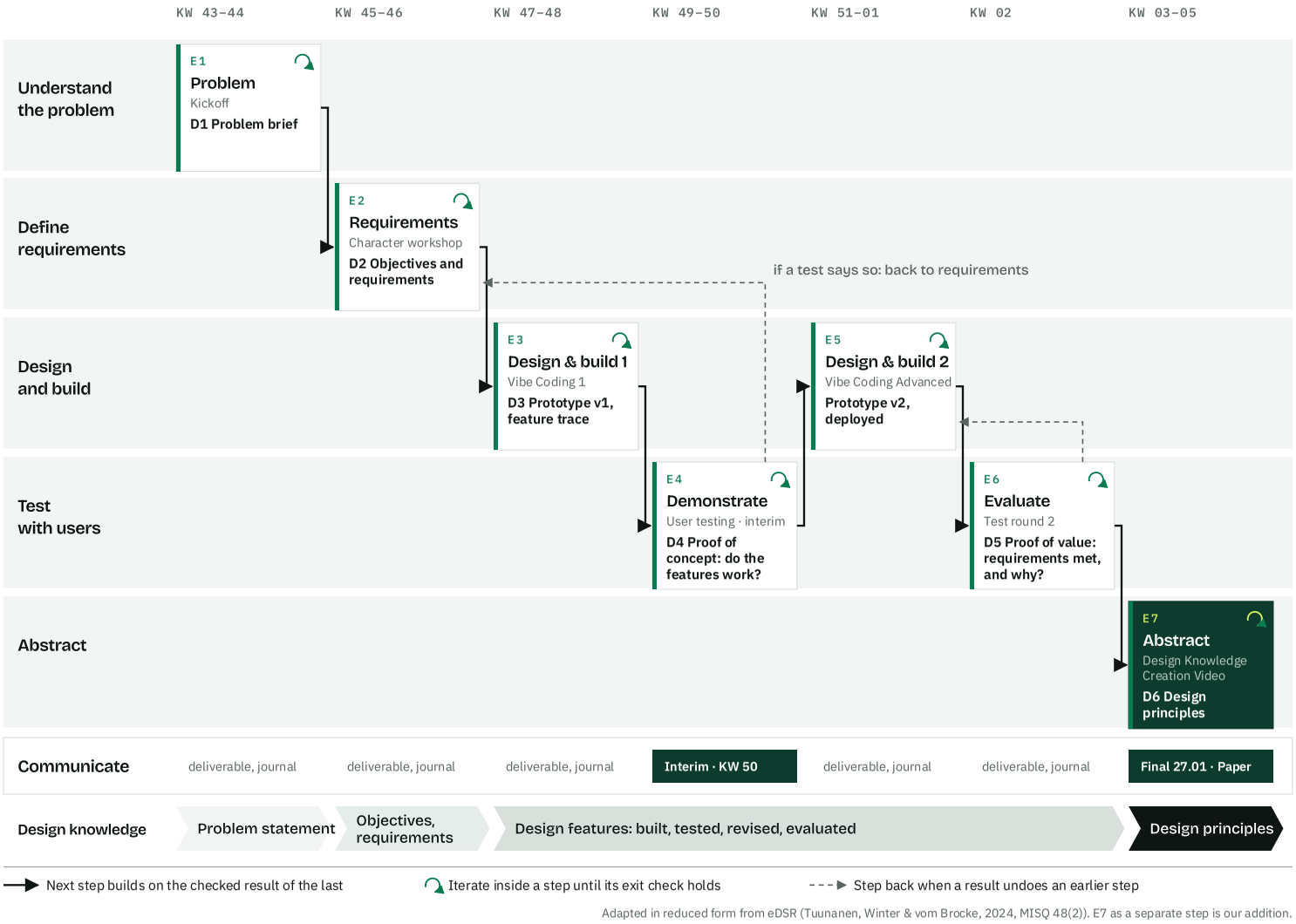
Five cases: each pair takes one

At the kickoff you visit five case stations, form a pair and hand in your top two cases. The full briefs are on the companion site.

	CASE	FOR WHOM	MODALITY	YOU GET	SUPERVISOR
1	**FinBot** How can a bank's service bot offer a really good user experience?	Bank customers and prospective customers; you focus on one pain point or user group	Chat, voice or avatar: your decision	A benchmark study with bots to try, printouts of Sparkassen bots, insights from the FinBot project in a meeting with Amilia	Amilia Fröhlich
2	**Personal finance** How can an agent help people understand their money across all their accounts and apps, without selling them anything?	People who manage money across several accounts and apps; you pick one group	Your decision; text with simple charts is a start	Synthetic personas with exports from several sources, a starter repository, last winter's seminar report and interview guide	Hans-Henning Näscher
3	**VIVA** When there is no driver, who reassures, informs and helps the passengers?	Passengers of autonomous public transport: everyone who boards, not only students	Voice, avatar or something else: your decision	Two core papers, the video of the earlier VIVA prototype, access to the campus experiment	Amilia Fröhlich
4	**Companion** People call it my Replika. How do we design a companion that people value without making them dependent on it?	Suggested: first-semester and international students in Münster in their first months	Your decision; text, voice optional	Three core papers, the online appendix of our companion study, apps to compare	Timo Strohmann
5	**Second Thought** Online shopping is built for one click. AI shopping assistants now advise us and even buy for us. Can an agent help people make more sustainable buying decisions, without patronising them?	Anyone who buys online; people who do not care about sustainability are the hardest and most interesting group	Your decision: advisor, companion over time or an agent that buys for the user; text, voice or other	Core papers, a mock product page from the kickoff station, the summary of our own study	Hans-Henning Näscher

METHOD · DESIGN SCIENCE, LIGHT

Seven steps, two rounds of build and test, one chain of design knowledge



Adapted in reduced form from eDSR (Tuunanen, Winter & vom Brocke, 2024, MISQ 48(2)). E7 as a separate step is our addition.

METHOD

Design science, light: seven steps

A project is a sequence of self-contained steps (echelons, Tuunanen et al., 2024). Each step ends with a result that passed its exit check. You design and build twice and test twice. Inside a step you iterate until the check holds; if a later result shows an earlier step was wrong, you step back. The figure on the previous page shows the whole semester.

	STEP	RESULT	EXIT CHECK
E1	Problem Understand the problem	D1 Problem brief	The problem statement names no technology. Each perspective rests on at least one concrete source: a document, a paper, a user. You can say what surprised you in the user conversations.
E2	Requirements Define requirements	D2 Character, objectives and requirements	Every requirement serves at least one objective. Every requirement is traced to a user insight or a theory. No requirement prescribes a technology or a specific feature.
E3	Design and build 1 Design and build	D3 Prototype v1 and feature trace	The prototype runs end to end for your core scenario. Every feature is linked to a requirement, and every requirement has at least one feature. You can show a conversation where the agent fails, and what happens then.
E4	Demonstrate Test with users	D4 Proof of concept (test round 1)	You tested with at least four users from outside the course. Each finding is linked to a feature and a requirement. Each decision for v2 follows from a finding.
E5	Design and build 2 Design and build	Prototype v2, deployed	Every change traces back to a finding from round 1. Test users can reach the prototype without your help. Logging captures what you need for round 2.
E6	Evaluate Test with users	D5 Proof of value (test round 2)	Every requirement has a verdict with evidence. Every verdict has a reason why. The limits of your data are written down.
E7	Abstract Abstract	D6 Design principles and final submission	Every principle rests on at least one evaluated feature and its reason why. A designer outside your team understands each principle without your paper. The paper states what the principles are grounded in, and what not.

The chain of design knowledge

01 **Problem statement** E1
What does not work, for whom, and why it matters, seen from practice, research and users.

02 **Design objectives and requirements** E2
Objectives say what the agent should achieve; requirements say what it must do or have to get there.

03 **Design features** E3 to E6
The concrete, buildable realisation of each requirement: built, tested with users, revised and evaluated.

04 **Design principles** E7
Abstracted from the evaluated features: what to do, for whom, in which context, and why it works (Gregor et al., 2020).

TWO TEST ROUNDS

E4 Demonstrate: do the features work for users? Formative, proof of concept. **E6 Evaluate:** are the requirements met, and why? Summative, proof of value. The reason why is what you abstract into design principles in E7.

DESIGN JOURNAL AND AI USE

After every step and every week: what we decided, because of what, what we doubt, and which AI tools we used for what (W10). The journal is part of your workbook submission.

E1

STEP E1 · KW 43–44 · UNDERSTAND THE PROBLEM · PROBLEM STATEMENT

Problem

Whose problem is worth an agent, and how do practice, research and users see it?

At the kickoff you visit five case stations, form a pair and hand in your top two cases. Each case comes with a roughly framed gap and material: papers, practice insights, sometimes user interviews. Then you look at the problem from three perspectives and talk to first users.

SESSIONS

Kickoff · Timo Strohmann · Tue 20.10.2026, 13:00 to 16:00. Input, case gallery walk, pairs and case choice. Understand what conversational agents are, why designing them is a research problem, and how we work with design science. Visit the five case stations, form a pair and hand in your top two cases.

YOU WILL BE ABLE TO

- Explain what distinguishes chatbots, assistants, companions and agents
- Name the four theory families we build on
- Describe your case's problem from the perspective of practice, research and users
- Talk to at least three potential users

DELIVERABLE D1 · END OF KW 44 (SUN 01.11), BEFORE THE CHARACTER WORKSHOP

Problem brief

One to two pages: the problem statement from the perspectives of practice, research and users; first user insights; your design question.

KEY CONCEPTS

Three perspectives. Practice: who runs such an agent and what goes wrong today. Research: what we know and what is open. Users: what people need and say. A good problem statement holds all three.

Socio-technical system. An agent is never only a model. What matters is the fit between user, task and technology (Maedche et al., 2019).

Time horizon. Ad-hoc supporter, temporary advisor or persistent companion: how long the relationship lasts changes almost every design decision (Nißen et al., 2021). Theory Lab: Time horizon.

Agency level. How much the agent decides and does on its own, and where the human stays in the loop (Holldack et al., 2026). Theory Lab: Agency dial.

TASKS

- ☐ At the kickoff: visit the case stations, form a pair and hand in your top two cases
- ☐ Read the case brief and the core papers of your case
- ☐ Try at least three existing agents in your domain
- ☐ Talk to at least three potential users (worksheet W2)
- ☐ Write the problem brief from three perspectives (worksheet W1)
- ☐ Start your design journal (worksheet W10)

WATCH OUT

- Starting from a technology (“we want to use RAG”) instead of a problem
- Only reading, never talking to users
- Assuming an agent is the right answer; sometimes a form is better

EXIT CHECK

- ☐ The problem statement names no technology.
- ☐ Each perspective rests on at least one concrete source: a document, a paper, a user.
- ☐ You can say what surprised you in the user conversations.

READING

Core. Diederich et al. (2022); Maedche et al. (2019).

Further. Schöbel et al. (2024); Følstad et al. (2021); Nißen et al. (2021).

W1 Problem brief: three perspectives

Describe the problem of your case before any solution, from the perspectives of practice, research and users. This becomes deliverable D1.

Team and case

The problem in one sentence
Who struggles with what, in which situation? No technology in this sentence.

Practice
Who runs or would run such an agent? What goes wrong today? Which rules and constraints apply?

Existing solutions we tried
At least three agents or services: what works, what fails

Research
What do we know (two to four sources)? What is still open or contested?

Users
Who are they, who else is affected, what did they tell us (W2)?

Why a conversational agent at all?
What would a form, a website or a human do better?

Our design question
How might we design an agent that … for … so that …?

W2 Interview guide and insights

Plan your first conversations with users (at least three before D1, more in step E2) and capture what you learn in a comparable way.

What we want to learn

Three research questions for the interviews

Opening questions

Context, routines, last time the problem occurred

Deepening questions

Ask for stories, not opinions: “Tell me about the last time …”

Questions about agents

Experiences with chatbots or assistants, expectations, concerns

INTERVIEW	WHO (ROLE, CONTEXT)	KEY QUOTE	NEED OR PAIN POINT	SURPRISE

W10 Design journal and AI transparency

Keep one short entry after every step and every week of team work. At the end, summarise your AI use in the transparency statement. Both are part of your workbook submission. W10 runs through all seven steps; it is printed here because you start it in E1.

DATE	WHAT WE DECIDED	BECAUSE (INSIGHT, SOURCE, TEST)	WHAT WE DOUBT	AI TOOLS USED, FOR WHAT, HOW CHECKED
03.11.	Companion never initiates after 22:00	Two interviewees felt pressured by late messages	Is this too paternalistic?	Claude: interview synthesis; checked against our notes

First row: example.

AI transparency statement: tools and versions

For which parts (code, analysis, text, images), and to what extent

How we checked the results

What we would not delegate to AI again, and why

E2

STEP E2 · KW 45–46 · DEFINE REQUIREMENTS · DESIGN OBJECTIVES AND REQUIREMENTS

Requirements

What should the agent achieve, and what must it do to get there?

In Amilia's three-hour workshop you build a persona from your interviews and develop your agent's character in depth. From problem and character you derive design objectives and requirements.

SESSIONS

Character and problem space · Amilia Fröhlich · KW 45 (02.11 to 06.11), day open. Workshop, 180 minutes: input, persona, character design, peer check. Deepen your understanding of the problem, develop a thought-through agent character, and derive traceable design objectives and requirements from it.

YOU WILL BE ABLE TO

- Design the agent's character with the social cue taxonomy
- Formulate three to five design objectives
- Derive design requirements and trace each to a user insight or to theory

DELIVERABLE D2 · END OF KW 46 (SUN 15.11)

Character, objectives and requirements

Character sheet; three to five design objectives; design requirements, each traced to a user insight or to theory.

KEY CONCEPTS

Computers as social actors. People apply social rules to machines that show even minimal social cues. Neutral design does not exist, so every cue is a decision (Nass & Moon, 2000). Theory Lab: Cue mixer.

Social cues. 48 cues in four families: verbal, visual, auditory and invisible, such as response timing (Feine et al., 2019). Choose the cues you send, and the ones you deliberately do not send. Theory Lab: Cue mixer.

Anthropomorphism. Human-likeness raises expectations. Match name, face, voice and personality to what the agent can actually do (Epley et al., 2007; Seeger et al., 2018). Theory Lab: Uncanny valley.

Persona and character. The persona describes the user; the character describes the agent. Both are design decisions, and the character follows from the persona's needs.

Objective and requirement. An objective is an intended effect for users (“students stay in control”). A requirement is what the agent must do or have to reach it (“users can see and delete what is remembered”).

TASKS

- ☐ Bring your problem brief to the workshop
- ☐ Fill the character sheet (worksheet W3)
- ☐ Formulate objectives and requirements (worksheet W4)
- ☐ Update your design journal

WATCH OUT

- A persona that is a demographic profile without needs
- Giving the agent a human name and face without a reason
- Requirements that already describe features (“a chat bubble with emojis”)

EXIT CHECK

- ☐ Every requirement serves at least one objective.
- ☐ Every requirement is traced to a user insight or a theory.
- ☐ No requirement prescribes a technology or a specific feature.

READING

Core. Feine et al. (2019); Nißen et al. (2021).

Further. Nass & Moon (2000); Seeger et al. (2018); Ribeiro et al. (2024); Clark et al. (2019); Pentina et al. (2023); Ahmad et al. (2021).

W3 Persona and character sheet

Define who the agent is for, and who the agent is. Every character choice needs a reason from your research or from theory.

Primary persona
Name, situation, goals, frustrations, relevant quote from an interview

Journey moment where the agent helps

Agent role
Expert, facilitator or peer? Assistant, advisor or companion?

Modality and embodiment (text, voice, avatar), and why

Personality in three adjectives, and why

Time horizon (Nißen et al., 2021)

☐ Ad-hoc supporter ☐ Temporary advisor ☐ Persistent companion

Agency level (Holldack et al., 2026)

☐ Assisting ☐ Collaborative ☐ Autonomous with human on the loop

CUE FAMILY	CUES WE SEND	CUES WE DELIBERATELY DO NOT SEND	REASON (RESEARCH OR THEORY)
Verbal	Informal, short sentences	Emoji	Users are stressed; clarity first

First row: example.

How the agent discloses that it is an AI (EU AI Act, Art. 50)

What the agent never does
Boundaries, handover rules

Workshop results

Paste photos of your sticky notes and posters from the character workshop, one box per part.

Persona

Life foundation: why the agent exists, for whom, what it knows

Personality: traits, tone, virtues and limits

Appearance: name, modality, social cues

Path and conflict: critical situations, what it never does, handover

W4 Objectives and requirements

Derive design objectives and design requirements from your problem and your users. This table grows through the semester; it is the backbone of D2.

Design objectives: what should the agent achieve for users?
Three to five, each an intended effect, e.g. “students feel supported and stay in control”

INSIGHT OR THEORY (SOURCE)	OBJECTIVE IT SERVES	DESIGN REQUIREMENT: WHAT THE AGENT MUST DO OR HAVE	RATIONALE
Users felt watched by apps that remember everything (I2, I4)	O2 Users stay in control	R3 The user can see and delete what the agent remembers	Trust calibration (Amershi et al., 2019)

First row: example.

- [ ] Every requirement serves at least one objective
- [ ] Every requirement has a source: a user insight or a theory
- [ ] No requirement prescribes a technology or a specific feature

E3

STEP E3 · KW 47–48 · DESIGN AND BUILD · DESIGN FEATURES

Design and build 1

Which features meet your requirements, and do they run?

In Hans's first vibe coding session you see how agents are built today and instantiate your character as a working prototype, starting with the system prompt. AI coding tools are welcome. For every requirement you design one or more features, and you keep the trace between them.

SESSIONS

Vibe Coding 1 · Hans-Henning Näscher · KW 47 (16.11 to 20.11), day open. 30 · 30 · 30: input, live case, own project. See how agents are built today and turn your requirements into design features of a first working prototype.

YOU WILL BE ABLE TO

- Design features for each requirement
- Build prototype v1 in the agreed tool stack
- Write and run test conversations, including ones that should fail gracefully

DELIVERABLE D3 · END OF KW 48 (SUN 29.11), BEFORE THE USER TEST SESSION

Prototype v1 and feature trace

A running agent, the build sheet, and the trace table: every feature linked to the requirement it serves.

KEY CONCEPTS

Design feature. A concrete, implementable property of the agent, for example a memory of the user's goals or a recap at the start of a session. Every feature serves a requirement.

Persona by prompt. The system prompt is where your character sheet becomes behaviour: role, tone, boundaries, handover rules.

Grounding and repair. Good agents confirm understanding, ask when things are ambiguous and recover from misunderstandings (Clark, 1992). Theory Lab: Grounding and repair.

Workflows before agents. Start with the simplest architecture that works and add autonomy only when needed (Schluntz & Zhang, 2024). Theory Lab: Agency dial.

TASKS

- ☐ Map requirements to features (worksheet W5)
- ☐ Fill the build sheet: prompt, memory, tools, guardrails (worksheet W6)
- ☐ Build prototype v1
- ☐ Write ten test conversations
- ☐ Log your AI tool use in the design journal (W10)

WATCH OUT

- Building features no requirement asks for
- Choosing a large model that answers too slowly for a conversation
- Putting real personal data into prompts or logs

EXIT CHECK

- ☐ The prototype runs end to end for your core scenario.
- ☐ Every feature is linked to a requirement, and every requirement has at least one feature.
- ☐ You can show a conversation where the agent fails, and what happens then.

READING

Core. Schluntz & Zhang (2024); Weisz et al. (2024).

Further. Park et al. (2023); Wang et al. (2024); Zamfirescu-Pereira et al. (2023); Strohmann et al. (2023).

W5 Feature trace

Design one or more features for every requirement and keep the trace. Update it in design and build 2; it becomes the feature table of your paper.

REQUIREMENT	DESIGN FEATURE	HOW IT IS BUILT (PROMPT, MEMORY, TOOL, INTERFACE)	VERSION (V1, V2) AND WHY CHANGED
R3 The user can see and delete what the agent remembers	Memory panel with delete and “forget this conversation”	JSON store; panel in the chat sidebar; system prompt cites only stored facts	v2: grouped by topic after round 1 (users could not find items)

First row: example.

- [ ] Every requirement has at least one feature
- [ ] Every feature serves a requirement
- [ ] Every change in v2 traces to a finding from round 1 (W8)

W6 Build sheet

Document how your agent is built, so that design features, tests and limits stay visible.

Tool stack and model

System prompt: role, persona, rules (summary)

Memory: what is remembered, for how long, and how the user can see or delete it

Tools and actions the agent can use

Knowledge sources (retrieval)

Guardrails and handover to a human

TEST CONVERSATION	EXPECTED BEHAVIOUR	OBSERVED BEHAVIOUR	FIX

E4

STEP E4 · KW 49–50 · DEMONSTRATE (TEST ROUND 1) · DESIGN FEATURES, TESTED

Demonstrate

Do your features work for real users?

In Amilia's user test session you plan both test rounds and pilot the first with another team. Round 1 is a demonstration: you show that the features work for users, and you learn what to change. The result is a proof of concept. Round 1 runs in KW 49 and 50, and you present the result at the interim presentations in KW 50. Start recruiting early: finding test users in December is hard.

SESSIONS

User testing · Amilia Fröhlich · KW 49 (30.11 to 04.12), day open. 90 minutes: input, demonstration, think-aloud exercise, peer pilot test. Learn to run a think-aloud test with a short interview, pilot your test plan with another team, and plan both test rounds.

Interim presentations · All · KW 50 (07.12 to 11.12), day open. Ten minutes talk with live demo, ten minutes discussion per team. Show your proof of concept after the first build and test, and collect feedback for design and build 2.

YOU WILL BE ABLE TO

- Plan two test rounds that link observations to features and requirements
- Run think-aloud sessions and short interviews
- Decide what to change for prototype v2

DELIVERABLE D4 · KW 50, WITH THE INTERIM PRESENTATION

Proof of concept (test round 1)

Test plan, findings per feature and requirement, decisions for prototype v2.

KEY CONCEPTS

Proof of concept. Round 1 shows that the design works for users in a realistic setting. It is formative: its purpose is to improve the design (Tuunanen et al., 2024).

Think-aloud. Users speak their thoughts while using the agent. You see where the design breaks, not only whether users like it.

Measures per requirement. For every requirement, decide before the test what you observe or ask, and what “met” and “not met” look like. Otherwise every result looks like success.

Informed consent. Participants know the purpose, what is recorded, how long it is stored and that they can stop at any time. Adapt the shared consent template and have it confirmed by Amilia.

TASKS

- ☐ Write the test plan for both rounds and the consent form (worksheet W7)
- ☐ Pilot round 1 with another team in the session
- ☐ Test with at least four external users
- ☐ Log findings and decisions (worksheet W8)
- ☐ Present at the interim presentations (KW 50)

WATCH OUT

- Testing only with fellow students
- Asking “Do you like it?” instead of observing
- Collecting data without informed consent

EXIT CHECK

- ☐ You tested with at least four users from outside the course.
- ☐ Each finding is linked to a feature and a requirement.
- ☐ Each decision for v2 follows from a finding.

READING

Core. Amershi et al. (2019); Tuunanen et al. (2024).

Further. Nielsen (2012); Benbasat & Wang (2005); Elshan et al. (2022).

W7 Test plan for both rounds

Plan round 1 (demonstration: do the features work for users?) and round 2 (evaluation: are the requirements met, and why?). Part of D4; reused for D5. Round 2 may start in KW 01 if prototype v2 is deployed early.

Round 1 (KW 49–50): participants, setting, tasks for the think-aloud

At least four external users

Round 2 (KW 02): participants, setting, tasks

At least three external users, at least one new to your prototype, ideally with the deployed prototype in their own setting

Interview questions after the task

Ask for experiences, not ratings: “What did you think the agent knew about you?” Use the interview guide from the user test session.

Roles and recording

Who moderates, who takes notes, audio yes or no (with consent)

REQUIREMENT	FEATURE	WHAT WE OBSERVE OR ASK	MET LOOKS LIKE	REASON QUESTION
R3 User can see and delete memory	Memory panel	Think-aloud while using the panel; interview on control	Users find, read and edit items; say they feel in control	What made you feel in control, or not?

First row: example.

- [ ] Consent form adapted from the shared template and confirmed by Amilia
- [ ] No personal data enters prompts or logs without consent
- [ ] Round 1 piloted with another team
- [ ] Round 2 dates booked before the break
- [ ] Data storage and deletion planned

W8 Results and the reasons why

Turn findings into decisions (round 1) and verdicts with reasons (round 2). Verdict in round 1: works, partly, does not; in round 2: met, partly, not met. Keep what users said apart from your own explanation: the explanations are the raw material for your design principles.

ROUND	FEATURE (V1/V2) AND REQUIREMENT	FINDING AND EVIDENCE (HOW MANY, WHICH DATA)	VERDICT	WHAT USERS SAID (QUOTE, PARTICIPANT)	OUR EXPLANATION (MECHANISM)	DECISION
2	Memory panel v2 / R3	5 of 6 users edited items	Met	“I could see what it keeps, so I trusted it” (P4)	Seeing the memory mattered more than how much was stored	Keep; principle candidate

First row: example.

What our data cannot show

E5

STEP E5 · KW 51–01 · DESIGN AND BUILD · DESIGN FEATURES, REVISED

Design and build 2

What changes because of what users showed you?

In Vibe Coding Advanced you take the prototype towards deployment: hosting, logging for the evaluation, guardrails and cost. You revise the features that round 1 questioned, and only those. If round 1 showed that a requirement was wrong, you step back and change it first.

SESSIONS

Vibe Coding Advanced · Hans-Henning Näscher · KW 51 (14.12 to 18.12) or KW 02 (11.01 to 15.01). Clinic: bring your problems. Revise your features and take the prototype towards deployment: hosting, logging for the evaluation, guardrails, cost.

YOU WILL BE ABLE TO

- Revise features based on the findings of round 1
- Deploy prototype v2 so that test users can reach it
- Set up logging for the evaluation

RESULT · END OF KW 01 (SUN 10.01)

Prototype v2, deployed

A deployed agent with logging; every change traced to a finding from round 1.

KEY CONCEPTS

Step back when needed. A test can show that a requirement, not the feature, was wrong. Then you go back to E2, change the requirement and record why in your journal (Tuunanen et al., 2024).

Deployment. A prototype users can reach on their own device, with logging you can analyse and costs you can control.

TASKS

- ☐ Revise features and update the trace (W5)
- ☐ Deploy prototype v2
- ☐ Prepare round 2 with the test plan (W7)

WATCH OUT

- Adding features instead of fixing what users struggled with
- Changing everything at once, so round 2 cannot tell what helped

EXIT CHECK

- ☐ Every change traces back to a finding from round 1.
- ☐ Test users can reach the prototype without your help.
- ☐ Logging captures what you need for round 2.

READING

Core. Schluntz & Zhang (2024).

Further. Weisz et al. (2024).

E6

STEP E6 · KW 02 · EVALUATE (TEST ROUND 2) · DESIGN FEATURES, EVALUATED

Evaluate

Are the requirements met, and why?

Round 2 is the evaluation: do the revised features meet the requirements, for users in their own setting? The result is a proof of value. For every feature you record not only whether it worked, but why. That “why” is what you will abstract into design principles in E7.

YOU WILL BE ABLE TO

- Evaluate whether each requirement is met
- Explain why each feature worked or did not
- Say what your data cannot show

DELIVERABLE D5 · END OF KW 02 (SUN 17.01)

Proof of value (test round 2)

Per requirement: met, partly or not met, with evidence and the reason why; limits of the data.

KEY CONCEPTS

Proof of value. Round 2 shows whether the design delivers what it should, for users in their own context. It is summative: its purpose is to judge the design (Tuunanen et al., 2024).

Reason questions. Questions that ask for the cause, such as “What made you feel in control, or not?”. “Users felt in control” is a finding; “because they could see what was remembered, not because little was remembered” is a mechanism. Mechanisms become principles.

TASKS

- ☐ Run round 2 with at least three external users, at least one of them new to your prototype (start in KW 01 if v2 is ready)
- ☐ Debrief ten minutes after each test: observation, what the user said, your explanation (W8)
- ☐ Use Amilia's open consultation slot in KW 02 if you need advice
- ☐ Write down what your data cannot show

READING

Core. Tuunanen et al. (2024); Iivari et al. (2021).

Further. Elshan et al. (2022); Benbasat & Wang (2005).

WATCH OUT

- Reporting satisfaction scores without reasons
- Claiming more than a handful of users can show
- Testing over the holidays without a plan

EXIT CHECK

- ☐ Every requirement has a verdict with evidence.
- ☐ Every verdict has a reason why.
- ☐ The limits of your data are written down.

E7

STEP E7 · KW 03–05 · ABSTRACT AND COMMUNICATE · DESIGN PRINCIPLES

Abstract

What does another designer learn from your work?

From the evaluated features and their reasons you formulate a small set of design principles in the anatomy of Gregor et al. (2020) and check that others can reuse them; the Design Knowledge Creation Video (from KW 02) shows how. In KW 03 or 04, Rangina Ahmad from Parloa gives an industry input: a chance to hold your draft principles against practice. You present on 27 January and submit on 5 February. Strong papers can go to the WI 2027 Student Track.

SESSIONS

Design Knowledge Creation Video · Timo Strohmann · KW 02 (from 11.01.2027), self-paced. Video, about 20 minutes, with an exercise. Turn your evaluated features and the reasons why they worked into design principles that other designers can reuse.

Industry input: agents in practice · Rangina Ahmad (Lead AI Agent Architect, Parloa) · KW 03 or KW 04 (18.01 to 26.01), day open. Talk and discussion with a guest from industry (not a hands-on session). Hear how conversational agents are designed, built and run in industry, and hold your draft design principles against practice before the final presentation.

Final presentations · All · Wed 27.01.2027. Live demo, design rationale and evidence per team. Show your agent and what another designer can learn from it.

YOU WILL BE ABLE TO

- Formulate two to four design principles from your evaluated features
- Check them against the reusability criteria
- Present and write a short design science paper

DELIVERABLE D6 · FRI 05.02.2027

Design principles and final submission

Principle cards; seminar paper (WI template, about 12 pages); artifact (code and link); workbook with AI transparency statement.

KEY CONCEPTS

Design principle anatomy. Aim, implementer, user, context, mechanism, enactors and rationale (Gregor et al., 2020). Theory Lab: Principle builder.

Reusability. Accessible, important, novel and insightful, actable, effective: the minimum criteria for reusable principles (Iivari et al., 2021).

Grounded, not proven. Your principles are grounded in evaluated features. They are not tested as principles yet. Say so in the paper: that is honest and it is a contribution.

TASKS

- ☐ Watch the Design Knowledge Creation Video (from KW 02)
- ☐ Write two to four principle cards (worksheet W9)
- ☐ Bring your draft principles to the industry input (KW 03 or 04)
- ☐ Final presentation on Wed 27.01.2027
- ☐ Seminar paper in the WI template following W11
- ☐ Complete the workbook and the AI transparency statement (W10)
- ☐ Submit by Fri 05.02.2027

WATCH OUT

- Describing the prototype instead of the knowledge
- Principles only your team can understand
- Claiming more than your evaluation shows

EXIT CHECK

- ☐ Every principle rests on at least one evaluated feature and its reason why.
- ☐ A designer outside your team understands each principle without your paper.
- ☐ The paper states what the principles are grounded in, and what not.

READING

Core. Gregor et al. (2020); Iivari et al. (2021).

Further. Gregor & Hevner (2013); Elshan et al. (2022); Strohmann et al. (2023); Peffers et al. (2007).

W9 Design principle card

Abstract one principle per card from your evaluated features and their reasons, in the anatomy of Gregor et al. (2020). The principle builder on the companion site helps you draft it.

Name of the principle

Aim: what the principle achieves or allows

Implementer and user

Context: where the principle applies, and where not

Mechanisms: what to do

Enactors: who or what acts

Rationale: why it works (theory, evidence)

Grounded in: which evaluated features and which reasons why (W8)

- [ ] Accessible: a practitioner understands it without our paper
- [ ] Important: it addresses a real problem users have
- [ ] Novel and insightful: it says more than common sense
- [ ] Actable: a developer knows what to do
- [ ] Effective: we have evidence that it works

W11 Seminar paper outline

A short design science paper in the WI conference template, about 12 pages. Use it as a checklist while writing. Strong papers can go to the WI 2027 Student Track.

- [ ] 1 Introduction: problem, relevance, research question, contribution
- [ ] 2 Background: conversational agents in your domain, theory you build on
- [ ] 3 Method: design science, light, in seven steps (Tuunanen et al., 2024); data sources
- [ ] 4 Problem, objectives and requirements: interviews, character, requirements
- [ ] 5 Design: features traced to requirements, prototype v1 and v2
- [ ] 6 Demonstration and evaluation: two rounds, verdict and reason per requirement
- [ ] 7 Design principles: in the anatomy format, with what they are grounded in
- [ ] 8 Discussion: contribution, implications for research and practice, limits
- [ ] 9 Conclusion
- [ ] Appendix: interview guide, test plan, feature trace
- [ ] Separate: AI transparency statement (W10)

APPENDIX A

Theory Lab

Seven simulated agents on the companion site. Each isolates one theory.

	LAB	THEORY	QUESTION	PATH
1	Cue mixer	Computers as social actors and social cues	What happens when you add or remove single social cues?	/lab/social-cues
2	Uncanny valley	Anthropomorphism	How human should your agent look and sound?	/lab/uncanny-valley
3	Grounding and repair	Common ground in conversation	What does it cost when an agent does not check what the user means?	/lab/common-ground
4	Trust calibration	Trust and acceptance	Can users tell when to rely on the agent?	/lab/trust
5	Time horizon	Relationships over time	How does an agent change when the relationship lasts months instead of minutes?	/lab/time-horizon
6	Agency dial	Agentic systems and delegation	How much should the agent do on its own?	/lab/agency
7	Principle builder	The anatomy of a design principle	Is your design principle something others could reuse?	/lab/dp-builder

APPENDIX B

Glossary

THE CHAIN OF DESIGN KNOWLEDGE

Problem statement (E1). What does not work, for whom, and why it matters, seen from practice, research and users. It names no technology.

Design objective (E2). An intended effect for users: what the agent should achieve. Objectives come from the problem statement and the persona.

Design requirement (E2). What the agent must do or have to reach an objective. It does not prescribe a technology or a specific feature, and it is traced to a user insight or a theory.

Design feature (E3 to E6). A concrete, buildable property of the agent that realises one or more requirements. Features are built, tested, revised and evaluated.

Design principle (E7). Prescriptive design knowledge abstracted from evaluated features: what to do, for whom, in which context, and why it works. Written in the anatomy of Gregor et al. (2020).

Feature trace (E3). The table that links every feature to the requirement it serves (worksheet W5). Every requirement has at least one feature; no feature exists without a requirement.

Reusability (E7). Whether others can apply a principle: accessible, important, novel and insightful, actable, effective (Iivari et al., 2021).

METHOD

Design science research (DSR). Research that creates and evaluates artifacts to solve a class of problems, and contributes design knowledge others can reuse. In this lab we use a light version in seven steps.

Step (echelon). A self-contained part of the project that ends with a checked result you can show (Tuunanen et al., 2024). The lab has seven: E1 to E7.

Exit check. The conditions under which a step is done. The next step builds on a result that passed its exit check, not on a draft.

Step back. When a later result shows that an earlier step was wrong (for example a requirement), you go back, fix it and record why in your design journal.

Persona (E2). A description of the user built from your interviews: situation, goals, frustrations, one real quote.

Character (E2). The designed identity of the agent: role, personality, appearance and how it handles critical situations. The character follows from the persona's needs.

Design journal. A short entry after every step and every week: what we decided, because of what, what we doubt, and which AI tools we used (worksheet W10).

TESTING AND EVALUATION

Proof of concept (E4). Test round 1: shows that the design works for users in a realistic setting. Formative: its purpose is to improve the design.

Proof of value (E6). Test round 2: shows whether the design delivers what it should, for users in their own context. Summative: its purpose is to judge the design.

Think-aloud (E4). Users say what they think while using the agent; you do not help or explain. You see where the design breaks, not only whether users like it.

Reason question (E6). A question that asks for the cause, such as “What made you feel in control, or not?”. Its answers turn findings into mechanisms.

Mechanism (E6 to E7). Why a feature works: “users felt in control because they could see what was remembered”. Mechanisms are what you abstract into design principles.

APPENDIX C

References

Adam, M., Wessel, M., & Benlian, A. (2021). AI-based chatbots in customer service and their effects on user compliance. Electronic Markets, 31(2), 427–445. https://doi.org/10.1007/s12525-020-00414-7

Ahmad, R., Siemon, D., Gnewuch, U., & Robra-Bissantz, S. (2021). The benefits and caveats of personality-adaptive conversational agents in mental health care. AMCIS 2021 Proceedings. https://aisel.aisnet.org/amcis2021/sig_hci/sig_hci/21

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. CHI 2019. https://doi.org/10.1145/3290605.3300233

Benbasat, I., & Wang, W. (2005). Trust in and adoption of online recommendation agents. Journal of the Association for Information Systems, 6(3), 72–101. https://doi.org/10.17705/1jais.00065

Brîncoveanu, C., Carl, K. V., & Hinz, O. (2026). Designing hyper-personalized financial co-pilots: An artifact for building trust through conversational advice. In DESRIST 2026, LNCS (pp. 167–184). Springer. https://doi.org/10.1007/978-3-032-28316-0_10

Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. Proceedings of the ACM on Human-Computer Interaction, 5(CSCW1), 188. https://doi.org/10.1145/3449287

Clark, H. H. (1992). Arenas of language use. University of Chicago Press.

Clark, L., Pantidi, N., Cooney, O., Doyle, P., Garaialde, D., Edwards, J., Spillane, B., Gilmartin, E., Murad, C., Munteanu, C., Wade, V., & Cowan, B. R. (2019). What makes a good conversation? Challenges in designing truly conversational agents. CHI 2019. https://doi.org/10.1145/3290605.3300705

Diederich, S., Brendel, A. B., Morana, S., & Kolbe, L. (2022). On the design of and interaction with conversational agents: An organizing and assessing review of human-computer interaction research. Journal of the Association for Information Systems, 23(1), 96–138. https://doi.org/10.17705/1jais.00724

Elshan, E., Engel, C., Ebel, P., & Siemon, D. (2022). Assessing the reusability of design principles in the realm of conversational agents. DESRIST 2022, LNCS 13229, 128–141. https://doi.org/10.1007/978-3-031-06516-3_10

Epley, N., Waytz, A., & Cacioppo, J. T. (2007). On seeing human: A three-factor theory of anthropomorphism. Psychological Review, 114(4), 864–886. https://doi.org/10.1037/0033-295X.114.4.864

Feine, J., Gnewuch, U., Morana, S., & Maedche, A. (2019). A taxonomy of social cues for conversational agents. International Journal of Human-Computer Studies, 132, 138–161. https://doi.org/10.1016/j.ijhcs.2019.07.009

Følstad, A., Araujo, T., Law, E. L.-C., Brandtzaeg, P. B., Papadopoulos, S., Reis, L., Baez, M., Laban, G., McAllister, P., Ischen, C., Wald, R., Catania, F., Meyer von Wolff, R., Hobert, S., & Luger, E. (2021). Future directions for chatbot research: An interdisciplinary research agenda. Computing, 103(12), 2915–2942. https://doi.org/10.1007/s00607-021-01016-7

Fröhlich, A. N., Gottschewski-Meyer, P., & Strohmann, T. (2026). Design features for virtual in-vehicle assistants in autonomous public transport. In HCI International 2026, LNCS 16741 (pp. 186–208). Springer. https://doi.org/10.1007/978-3-032-30427-8_12

Gnewuch, U., Morana, S., & Mädche, A. (2017). Towards designing cooperative and social conversational agents for customer service. In ICIS 2017 Proceedings. https://aisel.aisnet.org/icis2017/HCI/Presentations/1

Gnewuch, U., Morana, S., Adam, M. T. P., & Maedche, A. (2018). Faster is not always better: Understanding the effect of dynamic response delays in human-chatbot interaction. ECIS 2018 Proceedings. https://aisel.aisnet.org/ecis2018_rp/113/

Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. MIS Quarterly, 37(2), 337–355. https://doi.org/10.25300/MISQ/2013/37.2.01

Gregor, S., Chandra Kruse, L., & Seidel, S. (2020). Research perspectives: The anatomy of a design principle. Journal of the Association for Information Systems, 21(6), 1622–1652. https://doi.org/10.17705/1jais.00649

Hevner, A. R. (2007). A three cycle view of design science research. Scandinavian Journal of Information Systems, 19(2), 87–92. https://aisel.aisnet.org/sjis/vol19/iss2/4/

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. MIS Quarterly, 28(1), 75–105. https://doi.org/10.2307/25148625

Hildebrand, C., & Bergner, A. (2021). Conversational robo advisors as surrogates of trust: Onboarding experience, firm perception, and consumer financial decision making. Journal of the Academy of Marketing Science, 49(4), 659–676. https://doi.org/10.1007/s11747-020-00753-z

Holldack, F., Banh, L., & Strobel, G. (2026). Agentic information systems. Electronic Markets, 36, 5. https://doi.org/10.1007/s12525-025-00861-0

Iivari, J., Rotvit Perlt Hansen, M., & Haj-Bolouri, A. (2021). A proposal for minimum reusability evaluation of design principles. European Journal of Information Systems, 30(3), 286–303. https://doi.org/10.1080/0960085X.2020.1793697

Lewis, M., & Perry, M. (2019). Follow the money: Managing personal finance digitally. In Proceedings of CHI 2019 (pp. 1–14). ACM. https://doi.org/10.1145/3290605.3300620

Maedche, A., Legner, C., Benlian, A., Berger, B., Gimpel, H., Hess, T., Hinz, O., Morana, S., & Söllner, M. (2019). AI-based digital assistants: Opportunities, threats, and research perspectives. Business & Information Systems Engineering, 61(4), 535–544. https://doi.org/10.1007/s12599-019-00600-8

Mori, M., MacDorman, K. F., & Kageki, N. (2012). The uncanny valley. IEEE Robotics & Automation Magazine, 19(2), 98–100. (Original work published 1970) https://doi.org/10.1109/MRA.2012.2192811

Namyslo, N. M., Jung, D., & Sturm, T. (2025). The state of robo-advisory design: A systematic consolidation of design requirements and recommendations. Electronic Markets, 35, 20. https://doi.org/10.1007/s12525-025-00762-2

Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. Journal of Social Issues, 56(1), 81–103. https://doi.org/10.1111/0022-4537.00153

Nielsen, J. (2012). Thinking aloud: The #1 usability tool. Nielsen Norman Group. https://www.nngroup.com/articles/thinking-aloud-the-1-usability-tool/

Nißen, M., Selimi, D., Janssen, A., Rodríguez Cardona, D., Breitner, M. H., Kowatsch, T., & von Wangenheim, F. (2021). See you soon again, chatbot? A design taxonomy to characterize user-chatbot relationships with different time horizons. Computers in Human Behavior, 127, 107043. https://doi.org/10.1016/j.chb.2021.107043

Pan, S. L., Nishant, R., Tuunanen, T., & Choudrie, J. (2025). AI agents: Potential implications for IS research? Journal of Strategic Information Systems, 34(2), 101906. https://doi.org/10.1016/j.jsis.2025.101906

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. UIST 2023. https://doi.org/10.1145/3586183.3606763

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. Journal of Management Information Systems, 24(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Pentina, I., Hancock, T., & Xie, T. (2023). Exploring relationship development with social chatbots: A mixed-method study of Replika. Computers in Human Behavior, 140, 107600. https://doi.org/10.1016/j.chb.2022.107600

Regulation (EU) 2024/1689 of the European Parliament and of the Council (Artificial Intelligence Act), Article 50: Transparency obligations. https://artificialintelligenceact.eu/article/50/

Ribeiro, T., Veloso, A. I., & Brinson, P. (2024). Empathic characters for digital games: A prototype proposal. In HCI International 2024. Springer. https://doi.org/10.1007/978-3-031-60695-3_17

Schluntz, E., & Zhang, B. (2024, December 19). Building effective agents. Anthropic Engineering. https://www.anthropic.com/engineering/building-effective-agents

Schöbel, S., Schmitt, A., Benner, D., Saqr, M., Janson, A., & Leimeister, J. M. (2024). Charting the evolution and future of conversational agents: A research agenda along five waves and new frontiers. Information Systems Frontiers, 26(2), 729–754. https://doi.org/10.1007/s10796-023-10375-9

Schuetzler, R. M., Grimes, G. M., & Giboney, J. S. (2020). The impact of chatbot conversational skill on engagement and perceived humanness. Journal of Management Information Systems, 37(3), 875–900. https://doi.org/10.1080/07421222.2020.1790204

Seeger, A.-M., Pfeiffer, J., & Heinzl, A. (2018). Designing anthropomorphic conversational agents: Development and empirical evaluation of a design framework. ICIS 2018 Proceedings. https://aisel.aisnet.org/icis2018/hcri/Presentations/1/

Siemon, D., Strohmann, T., & Islam, N. (2026). Me and my Replika: Perceived affordances and the formation of psychological ownership of AI companions. European Journal of Information Systems. https://doi.org/10.1080/0960085X.2026.2673990

Skjuve, M., Følstad, A., Fostervold, K. I., & Brandtzaeg, P. B. (2021). My chatbot companion: A study of human-chatbot relationships. International Journal of Human-Computer Studies, 149, 102601. https://doi.org/10.1016/j.ijhcs.2021.102601

Staritz, J., Gottschewski-Meyer, P. O., Steuck, P.-F., Fröhlich, N. A., Strohmann, T., & Wartenberg, M. (2025). May I speak to the driver? Exploring design principles for a virtual in-vehicle assistance system applied in road-based autonomous public transport. In ECIS 2025 Proceedings. https://aisel.aisnet.org/ecis2025/hci/hci/1/

Strohmann, T., Siemon, D., Khosrawi-Rad, B., & Robra-Bissantz, S. (2023). Toward a design theory for virtual companionship. Human–Computer Interaction, 38(3–4), 194–234. https://doi.org/10.1080/07370024.2022.2084620

Takayanagi, T., Izumi, K., Sanz-Cruzado, J., McCreadie, R., & Ounis, I. (2025). Are generative AI agents effective personalized financial advisors? In Proceedings of SIGIR 2025 (pp. 286–295). ACM. https://doi.org/10.1145/3726302.3729897

Tuunanen, T., Winter, R., & vom Brocke, J. (2024). Dealing with complexity in design science research: A methodology using design echelons. MIS Quarterly, 48(2), 427–458. https://doi.org/10.25300/MISQ/2023/16700

Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., Chen, Z., Tang, J., Chen, X., Lin, Y., Zhao, W. X., Wei, Z., & Wen, J. (2024). A survey on large language model based autonomous agents. Frontiers of Computer Science, 18(6), 186345. https://doi.org/10.1007/s11704-024-40231-1

Weisz, J. D., He, J., Muller, M., Hoefer, G., Miles, R., & Geyer, W. (2024). Design principles for generative AI applications. CHI 2024. https://doi.org/10.1145/3613904.3642466

Wendt, M., Schaefer, S., & Schaarschmidt, M. (2025). Empowering regional bank sales through embodied conversational agents: A multiple case study. SIG SVC Pre-ICIS Workshop 2024. https://aisel.aisnet.org/sprouts_proceedings_sigsvc_2024/9

Zamfirescu-Pereira, J. D., Wong, R. Y., Hartmann, B., & Yang, Q. (2023). Why Johnny can't prompt: How non-AI experts try (and fail) to design LLM prompts. CHI 2023. https://doi.org/10.1145/3544548.3581388